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ON-FARM CONSERVATION PRIORITIES THROUGH A MULTICRITERIA MONO-SPECIFIC APPROACH

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Abbreviations:

ha, hectare; km, kilometre; t, tonne

ABSTRACT

Landraces are key plant genetic resources due to their adaptation to the local environment, which makes them likely to contain alleles or gene complexes that confer local adaptation.

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After massive loss of landraces in the past century, the diversity that still persists in farms is currently threatened and, thus, it must be safeguarded with the highest priority. On-farm conservation is an important management approach that allows landraces to evolve according to environmental changes. A methodological approach to systematic identification of the most appropriate areas (MAPA) for on-farm conservation of landraces is presented, using Spanish tomato landraces as a case study. Among a set of potential factors that might condition the selection of the MAPA, a panel of experts determined that the most important criteria in the assessment of the MAPA were related to agrobiodiversity and to ecogeographical parameters. For each important criterion, a specific 5x5 km resolution GIS layer was created, and six different prioritization strategies were applied on the area of study to obtain a final priority value per cell, and then an average priority value per county. The MAPA were assigned to the counties with the highest average priority values, according to the selected strategy. They were widely distributed in rural areas across the Iberian Peninsula, mainly in the autonomous communities of Valencia, the Basque Country, Extremadura, Northern Andalusia, Catalonia, Northern Aragon and Western Castilla-La Mancha. The methodological approach and priority counties identified could be used by governmental agencies to allocate *in situ* conservation projects.

INTRODUCTION

The conservation of plant genetic resources for food and agriculture (PGRFA) is essential to guarantee global food security in the current state of climate change and growing human population. Among plant genetic resources (PGR), landraces are key elements. Landraces, variable and identifiable populations lacking “formal” crop improvement, are specifically adapted to the environmental conditions of their cultivation area and to the traditional habits

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of the farmers who continue to grow them (Negri et al., 2009). However, they are currently one of the most threatened elements of agricultural PGR (Negri et al., 2009; Kell et al., 2012) and many studies have confirmed the effects of genetic erosion in different crops and regions (for instance, Hammer et al., 1996; Cebolla-Cornejo et al., 2007; Dyer et al., 2014; Martínez-Castillo et al., 2016; and Giupponi et al., 2020). In order to face those threats, landraces must be safeguarded with the highest priority (Negri et al., 2009) and their conservation reinforced.

Both the Convention on Biological Diversity (CBD, 1992) and the Second Global Plan of Action for PGRFA (FAO, 2011), as well as Sustainable Development Goal 2.5 (UN, 2015), have identified *ex situ* and *in situ* conservation of PGR as critical and complementary activities. Furthermore, the voluntary guidelines for the conservation and sustainable use of farmers' varieties/landraces (FAO, 2019a), indicate that *in situ* conservation of landraces should be complementary to *ex situ* conservation.

On-farm conservation is an important *in situ* conservation technique that allows the landrace to evolve according to the farmer's needs and to socio-cultural, economic and environmental changes (Brush, 2000). It aims at retaining potentially useful but undetermined genetic variation, and to generating new variation important to maintain the capacity of crops to adapt to changes (Bellon, 2009). However, since the first steps on exploration and promotion of on-farm conservation (during 80s-90s), there has been a significant criticism of the potential for this type of conservation due to the dynamism of agriculture and the lack of efficient ways to boost farmers' motivations to maintain landraces in the field (Frankel & Soule, 1981; Zeven, 1996; Peres, 2016).

Due to the high costs involved in on-farm conservation, conservation efforts should focus on areas where they lead to better impacts. Most recent studies on on-farm conservation priorities have been based on genetic studies to suggest where conservation efforts on

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landraces should be focused, such as Busso et al. (2000), Ben Romdhane et al. (2018), Lioi et al., (2019) and Rhoné et al. (2020). However, there are few standardized methods or protocols to identify areas appropriate for landrace conservation, and very few that take a geospatial approach. Most of these studies have proposed mono-specific or mono-genus methodological approaches (Tapia, 2015; Vera, 2018; Muccillo et al., 2019; Santillán-Fernández et al., 2021). In contrast, Paccico et al. (2018), focusing not only on landraces but also on crop wild relatives, showed a multi-specific methodological approach to identify agrobiodiversity hotspots for priority PGR *in situ* conservation. To the best of our knowledge, only Tapia (2015) and Vera (2018), who identified most appropriate areas (MAPA) for on-farm conservation of maize landraces in the mountains of Ecuador and of *Crataegus* spp. landraces in Mexico, respectively, have carried out studies on systematic geospatial identification of the MAPA, considering non-agroecological factors such as social or economic variables. And as far as we know, there have been no previous studies about on-farm conservation prioritization in Spain or in other European countries with the exception of Italy.

We selected tomato (*Solanum lycopersicum* L.) to gain further insight into the potential of mono-specific approaches (versus multi-specific ones, such as Paccico et al., 2018) in the identification of MAPA. Tomato is the vegetable with the highest values of production and harvested area in Spain [5 000 560 t, and 56 940 ha in 2019, respectively (FAO, 2019b)]. Furthermore, Spain is considered a secondary center of diversity of tomato (Bergougnoux, 2014) after its domestication in the Americas (Vavílov, 1992). Thus, the Spanish Inventory of Plant Genetic Resources (SIPGR, 2018) has more than 3000 accessions of tomato landraces collected all over the 50 Spanish provinces (see Spanish autonomous communities and provinces in Figure 1), and 68% of them have some characterization data

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available. We considered the compilation of social and economic data in this study, given their importance in a systematic planning of biodiversity conservation (Sarkar et al., 2011; Montesano et al., 2012; Bellon & van Etten, 2014; Tobón & Koleff, 2016; Wittman et al., 2017). The aim of this paper was to set a methodological approach to systematic identification of the most appropriate areas for on-farm conservation of Spanish tomato landraces. This kind of top-down approach can be useful in countries where GIS, georeferenced data and contribution of a panel of experts are available. By applying this approach, we aimed to reach the following objectives: (i) to identify the most important criteria to select the MAPA of tomato landraces in Spain, according to experts contributions; (ii) to identify the priority areas to focus on-farm conservation efforts on tomato landraces in Spain.

MATERIALS AND METHODS

Area of study

The area considered in this study was the territory where there is documented evidence of tomato landraces being grown in the present or in the past in Spain. We used as a proxy of these places the data available in the Spanish Inventory of Plant Genetic Resources (SIPGR, 2018) and other information about landraces still non-included in the SIPGR, provided in 2018 by the University Institute for Conservation and Breeding of Valencian Agrodiversity (COMAV). Within these data, accessions without geographic coordinates were excluded from the study, as well as accessions data with a georeferencing quality score below 60, according to GEOQUAL from CAPFITOGEN toolkit (Parra-Quijano et al., 2015). This

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procedure of evaluation of the georeferencing quality eliminates records with georeferencing data of low accuracy, mistaken coordinates or inconsistencies between coordinates and the associated locality names.

Subsequently, accessions of landraces with generic names or with identical local names in identical collection sites, that could induce errors in the estimation of landrace richness, were eliminated from the dataset according to the following criteria: (i) accessions without any local name or with a too generic one (such as *"Tomate"* or *"Tomate del país"*) were removed; (ii) when there were several accessions collected in the same place and with the same local name, they were considered duplicated samples and only one of them remained in the dataset, while the others were removed; (iii) accessions with more generic local names than other accessions collected in the same place were removed, when it was possible to consider that the accession with the more generic name could be a duplicate sample of one of the other accessions (i.e., *"Tomate grande rojo"* and *"Tomate grande rosa"* remained in the dataset whereas the accession *"Tomate grande"*, collected in the same place than the latter two, was discarded).

Finally, the local names of the remaining accessions were harmonized to estimate, in the most accurate way, the number of different landraces within an area. For example, "*Tomate cherry"*, *"Tomate cerecero"* and *"Tomate de cereza"* were homogeneized to *"Tomate cherry",* and *"Tomate de colgar", "Tomate de cuelga"* and *"Tomate de penjar"* were homogeneized to *"Tomate de colgar"*.

Using the final dataset, a 10 km radius buffer area around the collection site of each tomato landrace accession was created. The area defined by these buffer zones was considered the area of study and was subdivided into a grid of 5x5 km cells.

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Criteria selection for the identification of the most appropriate areas (MAPA) for on-farm conservation

A questionnaire was sent to 23 experts involved in tomato landraces (breeders, researchers, curators and leaders of non-governmental organizations involved in conservation and promotion of agrobiodiversity). Experts were asked to assign to each of the criteria listed in Table 1 a score ranging from 0 to 10 (0 = no importance, 10 = highest importance) concerning its relevance for the identification of the most appropriate areas for on-farm conservation. They were also asked to indicate the favorable status for each criterion, i.e., the end of the variable range associated with the most interesting values for conservation (for example, if an expert considered that the average population age is an important variable to be taken into account in the assessment of the suitability of a site for establishing on-farm conservation, they should indicate if it would be better for conservation to have a high or a low value of average population age).

Based on the replies of the experts, a mean value was obtained for each criterion. Only criteria with a mean score value ≥ 5.0 ("important criteria", IC) were finally considered in the evaluation of the suitability of an area to be prioritized for on-farm conservation. Mean score values of the IC were summed to obtain a total value, and the mean value of each IC was divided by this total value to get a weighted average importance value for each IC.

The sources of the variables related to the IC and the processes to estimate the values for the IC in the area of study are detailed in Supplemental Table S1 and Supplemental Text S1, respectively.

Generation of important criteria layers

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For each IC, a specific 5x5 km resolution GIS layer was created for the territory under study, using ArcGIS 10.3 software (ESRI, 2011). Values for each quantitative criterion were standardized as follows:

$y = (x-\mu)/\sigma$

where μ and σ are the mean value and the standard deviation of the data set, respectively. Then, standardized values were escalated to a range between 0 and 1 (0 = less favorable state, 1 = most favorable state, according to experts' responses), as follows:

$y = (x - x_{min}) / (x_{max} - x_{min})$

where x_{min} and x_{max} are the minimum and the maximum values of the data set, respectively.

Final value of priority per cell

To obtain a final priority value per cell, six prioritization strategies were applied, by combining different number of criteria and different types of contributions of each criterion to the final value:

- Strategy 16W: The 16 GIS criteria layers were summed according to the weighted average importance value calculated per criterion from experts' responses.
- Strategy 16S: The 16 GIS criteria layers were summed with the same importance value per criterion.
- Strategy 10W: The 10 GIS criteria layers of the 10 IC with the highest weighted average importance values, were summed according to the weighted average importance value calculated per criterion from experts' responses.
- Strategy 10S: The 10 GIS criteria layers of the 10 IC with the highest weighted average importance values, were summed with the same importance value per criterion.

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- Strategy 5W: The 5 GIS criteria layers of the 5 IC with the highest weighted average importance values, were summed according to the weighted average importance value calculated per criterion from experts' responses.
- Strategy 5S: The 5 GIS criteria layers of the 5 IC with the highest weighted average importance values, were summed with the same importance value per criterion.

The priority values per cell of each one of the six strategies were utilized to identify the MAPA for on-farm conservation of tomato landraces. Final priority values per cell were utilized to assign an average priority value per county. Each cell contributed to the priority value of the counties in which it was contained in proportion to its area within each county. In other words, counties were assigned priority values through an area weighted average of values across cells. Those counties with the highest average priority values were considered the MAPA.

RESULTS

Area of study

We compiled a total of 3064 and 114 (3178 total) records of tomato landraces from SIPGR and COMAV, respectively, collected between 1980 and 2014 (when the collection date was available). Seventeen accessions data with a georeferencing quality score below 60, according to GEOQUAL, were discarded in the study. After that, 774 and 456 (1247 total) accessions were removed from the data set due to lack of local name or having a too generic one, and for being considered duplicate samples of other accessions remaining in the dataset, respectively. Therefore, 1931 records of landraces were considered in the analysis. Their passport data are available in Supplemental Table S2. After creating the buffer area of 10 km

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radius around each collection site, the total area was subsequently subdivided into a grid of 9884 5x5 km cells, which comprised 47% of the Spanish territory.

Criteria selection for the identification of the most appropriate areas (MAPA) for on-farm conservation

Based on the responses of the expert survey, 16 out of 28 criteria were considered important criteria (IC) by experts and were therefore selected for the subsequent analysis. Their weighted average importance (from 0 to 1) ranged between 0.0512 and 0.0924 (Table 2). Phenotypic diversity of the tomato landraces and number of tomato landraces grown in the area were the IC with the highest scores in weighted average importance, followed by the number of species (other than tomato) grown in the area and the existence of environmental conditions that favour the cultivation of tomato landraces with potential adaptation to high temperatures and saline soils (Table 2).

Generation of important criteria layers

The values of the different variables related to the selected criteria varied across the territory, with the exception of Criterion 11 (Permission of tomato production with protected geographical indication), where all the cells in the area of study were assigned the value 0. Supplemental Table S3 provides the ranges and average values of the IC in the area of study. Maps with raw values of each one of the IC for the area of study are shown in Supplemental Figures S1-S4.

With regard to Criterion 2 (Number of other species grown with landraces in the area), a detailed analysis of the data in the area of study showed that tomato cultivation is

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accompanied in the cells, on average, by landraces of nine different horticultural crops (mainly bean, lettuce, pumpkin, melon, onion, broad bean, collard, garlic and cucumber, according to the data obtained from SIPGR), and these horticultural crops represent, on average, over 66% of the total crops with landraces grown in the area of study.

Final value of priority per cell

Using the 16 criteria and summing the scores with a weighted average according to the importance value provided by the experts (strategy 16W), values of conservation priority per cell ranged between 0.11 and 0.49 (Figure 2). Maps on final values of priority per cell for the rest of the strategies are shown in Supplemental Figures S5-S7. Autonomous community of Extremadura and some locations in Western Castilla-La Mancha, the Valencian Community, the Basque Country, Northern Aragon, Catalonia and Northern Andalusia contained a high number of cells with the highest scores (Figure 2). When the number of criteria involved in the calculation of the final value of priority per cell decreased, most priority cells grouped in Extremadura, Northern Andalusia, Southern Castilla-La Mancha and Northern Aragon, whereas the priority of the cells in the autonomous communities of the Basque Country, Catalonia and the Valencian Community, declined (Figure 2 and Supplemental Figures S5-S7).

After values per cell were aggregated by counties, the top ten priority counties scarcely differed when the strategies that match the number of criteria considered in obtaining the final value were compared (Supplemental Table S4). However, there were relevant differences among the top ten priority counties according to the number of criteria selected by the strategy. Similarly, the top ten cells of the six rankings of priority for on-farm conservation of tomato landraces had only one cell in common (Supplemental Table S4).

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Average non-standardized values per criteria of the top ten priority counties for on-farm conservation of tomato landraces in Spain are shown in Supplemental Tables S5-S10 for each of the strategies. Six and seven of the top ten priority counties for 16W strategy equaled or exceeded the average raw values in the area of study (shown in Supplemental Table S3) for the two most important criteria according to experts (i.e., for phenotypic diversity of tomato landraces and number of tomato landraces), respectively. In addition, nine of the top ten priority counties had lower average areas of the horticultural properties than the mean value in the area of study. In the top ten counties, average population age ranged between 38 and 52 years, and distance to an urban center of more than 10 000 inhabitants was lower than 35 km. Finally, all the top ten priority counties were totally or partially (>70% of the county area) included within the range of a "*Km0 Restaurant*", and over 90% of their area was declared as disadvantaged in eight of the top ten counties.

The MAPA, i.e., the counties with the highest average priority values, are shown in Figure 3 (for 16W strategy) and in Supplemental Figures S8-S10 (for the rest of the strategies). According to the results of applying the 16W strategy, the priority counties for on-farm conservation of tomato landraces in Spain are located in the autonomous communities of Valencia, the Basque Country, Extremadura, Northern Andalusia, Catalonia, Northern Aragon, Western Castilla-La Mancha and Murcia (Figure 3). The trend shown when analyzing final values of priority per cell was also observed when information was aggregated to the county level: when the number of criteria involved in the calculation of the final value of priority per cell decreased, priority counties grouped in Extremadura, Northern Andalusia, Southern Castilla-La Mancha and Northern Aragon (Figure 3 and Supplemental Figures S8-S10).

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DISCUSSION

We present a methodological approach to the systematic identification of the most appropriate areas (MAPA) for on-farm conservation, using tomato landraces as a case study. After determining the most important criteria in the assessment of the MAPA with the help of a panel of experts, average conservation priority values per county were obtained. Conservation actions focused on the priority counties identified, distributed in rural areas across the Iberian Peninsula, would contribute to preserve the genetic diversity of tomato landraces in Spain.

Methodological approach and criteria selection

Most recent studies on on-farm conservation priorities have been based on genetic studies to suggest where conservation efforts on landraces should be focused (Busso et al., 2000; Ben Romdhane et al., 2018; Lioi et al., 2019; Rhoné et al., 2020). Other authors have combined molecular, environmental and cultural parameters in the process (Muccillo et al., 2019), or have explored germplasm abundance and diversity indices not based on molecular markers of a certain group of PGR, along with environmental descriptors of the cultivation sites (Conversa et al., 2020a).

Following the suggestions by Sarkar et al. (2011) and Tobón et al. (2016), among others, who encourage to the consideration of social and economic variables in a systematic planning of biodiversity conservation, we considered these kinds of criteria in our study. We believe that the consideration of demographical and socioeconomical factors, besides agricultural, ecological and ecogeographical ones, may allow to better identify the MAPA where the invested efforts in on-farm conservation lead to better impacts. Factors such as the average population age or the distance to a relatively populated urban center (both of them

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considered important criteria by experts in this study) might determine the long-term permanence of a landrace in a particular area. In this sense, Tapia (2015) considered 13 criteria in the assessment of MAPA for on-farm conservation of maize landraces in the mountains of Ecuador, among which there was none related to the size of the horticultural properties or the proportion of the area dedicated to maize cultivation, nor any variable related to vulnerability to climate change. Vera (2018) submitted to consultation 22 criteria to identify MAPA for on-farm conservation of *Crataegus* spp. landraces in Mexico, being the "*cultural*" criteria the most abundant ones, mainly related to the uses of the species and the existence of traditional knowledge related to them. Ureta et al. (2013) concluded that, in terms of the distribution of particular maize races, social variables included in their analysis (average years of education and ethnic group) were not as overall good predictors in comparison to environmental data, but they were also important components of the set of factors that seem to be related to the presence of particular races. In our study, it is remarkable that only one category of criteria from the eight proposed to experts for evaluation (ecological interest and protected areas) had not finally a representation among the important criteria. The rest of the categories were represented through the selection of the important criteria, which suggests that it was appropriate to consider such a wide range of categories in the questionnaire to experts.

As we expected, experts concluded that two of the most important criteria in the assessment of the MAPA for on-farm conservation of tomato landraces in Spain were the number of tomato landraces and their phenotypic diversity. This result is in agreement with previous studies on the subject that identified landrace diversity as an important criterion to determine which farmers' landraces to conserve (FAO, 2019a) or to identify agrobiodiversity hotspots for priority *in situ* conservation of PGR (Paccico et al., 2018). It also agrees with the

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two most important criteria identified by Vera (2018) and with the second most important criterion identified by Tapia (2015), which was the number of landraces per farm. Our results also agree with those of Conversa et al. (2020a), where the higher richness and diversity of species, the abundance of landraces and the number of sites allowed them to highlight some important areas for on-farm conservation of vegetable landraces. In addition, the fact that the number of species (other than tomato) grown in the area was identified as the third most important criterion in the assessment of the MAPA, reinforces the potential utilization of this approach as an initial step for future multi-species MAPA approaches.

It is worthy to note that the percentage of the area dedicated to tomato cultivation (criterion 8), despite being considered as an important criterion by experts, was not in the first positions ranked by weighted average importance value. Experts considered that greater percentages of area dedicated to tomato cultivation would be favourable for on-farm conservation. However, real data gathered in this study showed that large areas dedicated to tomato cultivation might not be very compatible with growing landraces. Indeed, there were 39 counties with greater area dedicated to tomato cultivation than the top then priority counties according to 16W strategy. The highest values for criterion 8 corresponded to counties in the provinces of Badajoz and Almería, where intensive tomato cultivation takes place; whereas the counties with the largest number of tomato landraces were located in the provinces of Huesca, Vizcaya, Barcelona, Guipúzcoa and Alicante. In addition, nine of the top ten priority counties had lower average areas of horticultural properties than the mean value in the area of study, which suggests that there might not be intensive cultivation of tomato in these counties. The incorporation of local farmers and other local stakeholders to the panel of experts might have provided us a wider and more accurate perspective for the evaluation of important criteria.

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We would like to remark that through the selection of the favourable status of the important criteria, experts helped us to identify the MAPA for on-farm conservation according to their knowledge and experience. In this sense, experts were free to interpret if MAPA for on-farm conservation were the areas exposed to higher threats or those ones in which best impacts could be achieved, taking into account that these areas might not eventually coincide. In addition, in some of the criteria (criteria 8, 21 and 28) experts differed in their responses about the favourable status of the criteria. In those cases, we established that the favourable status was the one supported by most experts. The different perspectives that were found concerning what determines the best areas to carry out on-farm conservation highlight the complexity of the issue.

In this study we considered that a mono-specific approach might lead to a more accurate selection of the criteria influencing the suitability of the areas for on-farm conservation than a multi-species one. Nevertheless, the addition of results of other mono-specific analyses of other target crops could lead us to a broader and more complete prioritization of areas where on-farm conservation efforts could be carried out. Given that tomato cultivation is accompanied in the cells of the area of study, on average, by landraces of nine different horticultural crops, most priority areas for on-farm conservation of these horticultural crops, as well as other extensive crops, could be identified and considered together with the highest priority areas identified in this study.

Obviously, the approach presented here may be useful in countries or regions where GIS, geo-referenced data and a panel of experts are available. The main benefit of this approach is the possibility of carrying out larger scale studies, involving a varied typology of criteria and species. One potential downside that could be argued is the lack of acceptance from local farmers. However, one may think that extra support provided to farmers and areas

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prioritized from a multicriteria approach by governmental or non-governmental organizations is likely to be welcomed.

Priority areas to focus on-farm conservation

We chose the county as the administrative unit to show the resulting priority areas for on-farm conservation because this intermediate spatial unit between the province and the municipality has a uniform character from an agricultural point of view, and agricultural extension offices operate at this level. This issue is especially important when the results of the analysis are to be applied to the reality of the territory. Conversa et al. (2020a) used the landscape units, from the regional territorial landscape plan, and showed that this approach was effective in the identification of important areas for the maintenance of vegetable landraces in Puglia region (Italy). Other authors (Fischer & Lindenmayer, 2007; Wu, 2013; Wittman et al., 2017) have also considered agricultural landscapes as particularly useful units to understanding challenges related to biodiversity conservation and food security, due to its meaningfulness from both ecological and social-institutional perspectives.

Because there were relevant differences in the counties selected as MAPA according to the number of criteria selected by the strategy, we considered that the most accurate strategy to better identify the areas where on-farm conservation efforts should be focused was the most comprehensive, that is, the one that took into account the 16 important criteria. Although there was a scarce difference between 16W and 16S strategies' top ten cells, we considered that it was a better approach the one that gives different weight to each criterion depending on the importance value assigned by experts, i.e., 16W strategy. The priority counties identified according to this strategy, were located in the autonomous communities of Valencia, the Basque Country, Extremadura, Northern Andalusia, Catalonia, Northern

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Aragon, Western Castilla-La Mancha and Murcia, and could be used by national or regional governmental agencies to carry out landraces' *in situ* conservation projects. These projects should be focused on boosting economic and cultural motivations to maintain landraces in the field.

Strategies to continue on-farm conservation efforts carried out by farmers over centuries should be defined and implemented in Spain, involving genebanks, private sector and local farmers in a coordinated way.

Firstly, a Spanish Network for on-farm conservation and sustainable use of landraces should be established, by combining the knowledge of existing stakeholders with analyses using geographic information system technology (such us the present study). Field surveys would be useful to validate the results of the MAPA for on-farm conservation of tomato landraces, to check if the tomato landraces included in this study are still cultivated and have been cultivated for at least one generation (25-30 years), as in Conversa et al. (2020a). In addition, an accompanying systematic characterization study would benefit the fine-tuning of the MAPA identification. These characterization activities could be specially focused on fruit quality parameters (as in Carillo et al., 2019; and Conversa et al., 2020b) and on the investigation of molecular profiles of the tomato landraces (as in García-Martínez et al, 2006; Cortés-Olmos et al., 2015; Castellana et al., 2020; and Caramante et al., 2021). Then, the goal of boosting motivations to maintain landraces in the field could be achieved by actions such as enhancing the value of landraces products through the use of mark labeling, supporting local food supply systems based on landraces and developing food chains based on landraces (ECPGR, 2014). Biodiversity seed fairs, payment for environmental services, repatriation of landraces and their promotion inside protected areas could also help to support community-based conservation of landraces (de Haan, 2021).

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Secondly, efforts should be implemented to improve and promote best practices to ensure that the goals of conserving genetic diversity of landraces are met within the Spanish network. They could be sustained by creating closer alliances between *ex situ* and *in situ* resource managers.

Thirdly, it would be essential to enhance the use of conserved landraces and to influence policy to guarantee a governance structure to ensure the effective operation of the Spanish network of on-farm conservation.

Limitations of the study

In the absence of a real inventory of *in situ* maintained landraces in Spain, in the style of the First Inventory of *In situ* Maintained Landraces in Italy (Negri et al., 2013), we opted for a rigorous approach based on the data of the Spanish National Inventory of Plant Genetic Resources (SIPGR, 2018). A potential limitation of the use of these data about landraces could be the existence of spatial bias in collecting activities. In this sense, it may be argued that the spatial distribution of the accessions could reflect the intensity of collecting activities rather than the real distribution of the landraces. Certain autonomous communities such as Valencian Community and Aragon have historically carried out great efforts to collect and conserve landraces within its boundaries and in nearby locations. However, a different situation occurs in Extremadura, the Basque Country and Andalusia, where, instead of having collected tomato landraces by themselves, centres of agriculture of other communities (especially from Valencia, Aragon and Madrid), have collected there. In spite of these different situations, several counties of Extremadura, the Basque Country and Andalusia recorded high number of landraces. Therefore, and although a certain spatial bias could be noticed, our observations indicate it is not as intense as the bias that can be observed in wild

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plant species occurrences due to biased sampling, where adequate consideration of the potential bias is needed in the interpretation of the results (Yesson et al., 2007; Hortal et al., 2008; Boakes et al., 2010; Maldonado et al., 2015).

Another issue to take into consideration is that, as it is common in studies that do not handle genetic data, there is no certainty that all the landraces considered are genetically unique. As Maxted et al. (2013) recognized, we do not have the certainty to assert that two landraces with different names are genetically different. On the contrary, a high non-authenticity of old cultivars of lettuce in genebank collections was shown by van de Wouw et al. (2011) after comparing the DNA markers profiles (based on amplified fragment length polymorphisms) of accessions presumed to be identical, based on their cultivar names. We did not have enough characterization data in this study to discern if a landrace cultivated with one name in an area is the same one that other grown with a different name in another area, or even in the same area. The ideal situation would have been to have a great amount of characterization data, including genetic data. However, genetic data were not available for the accessions considered in this study. Therefore, a thorough process of analysis of the collection sites and the local names of tomato landraces collected in Spain was carried out to minimize the inconvenience caused by this issue. It was a conservative approach that removed from the initial data set of tomato landraces those without any local name and those with a high probability of being a duplicate sample of other accessions collected in the same place. Through this process, along with local names harmonization, we guaranteed that an account of unique landraces, as accurate as possible, could be conducted within each cell of the area of study.

Conclusion

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By using a methodological approach that considered a set of agricultural, ecological, ecogeographical, demographical and socioeconomical factors selected by a panel of experts, we found that the best criteria for the assessment of the most appropriate areas for on-farm conservation of tomato landraces in Spain were (i) the phenotypic diversity of tomato landraces, (ii) the number of tomato landraces grown in the area, (iii) the number of different crops (other than tomato) with landraces grown in the area, and (iv) the environmental conditions that favour the cultivation of tomato landraces with potential tolerance to high temperatures and saline soils. This identification allowed us to determine the priority areas to focus on-farm conservation efforts on tomato landraces, which are located in the autonomous communities of Valencia, the Basque Country, Extremadura, Northern Andalusia, Catalonia, Northern Aragon, Western Castilla-La Mancha and Murcia. Moreover, the results of this study could also be considered as a starting point for a future multi-species approach. The addition of results of other mono-specific analysis of other target crops, would lead us to a broader and more complete prioritization of areas where on-farm conservation efforts could be carried out. This methodology could be applied in other countries or regions. Further analyses could be carried out when other criteria, such as cultural value, farmers' priorities and threat status of the landraces, are available.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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FIGURE CAPTIONS

Figure 1. Spanish autonomous communities and provinces.

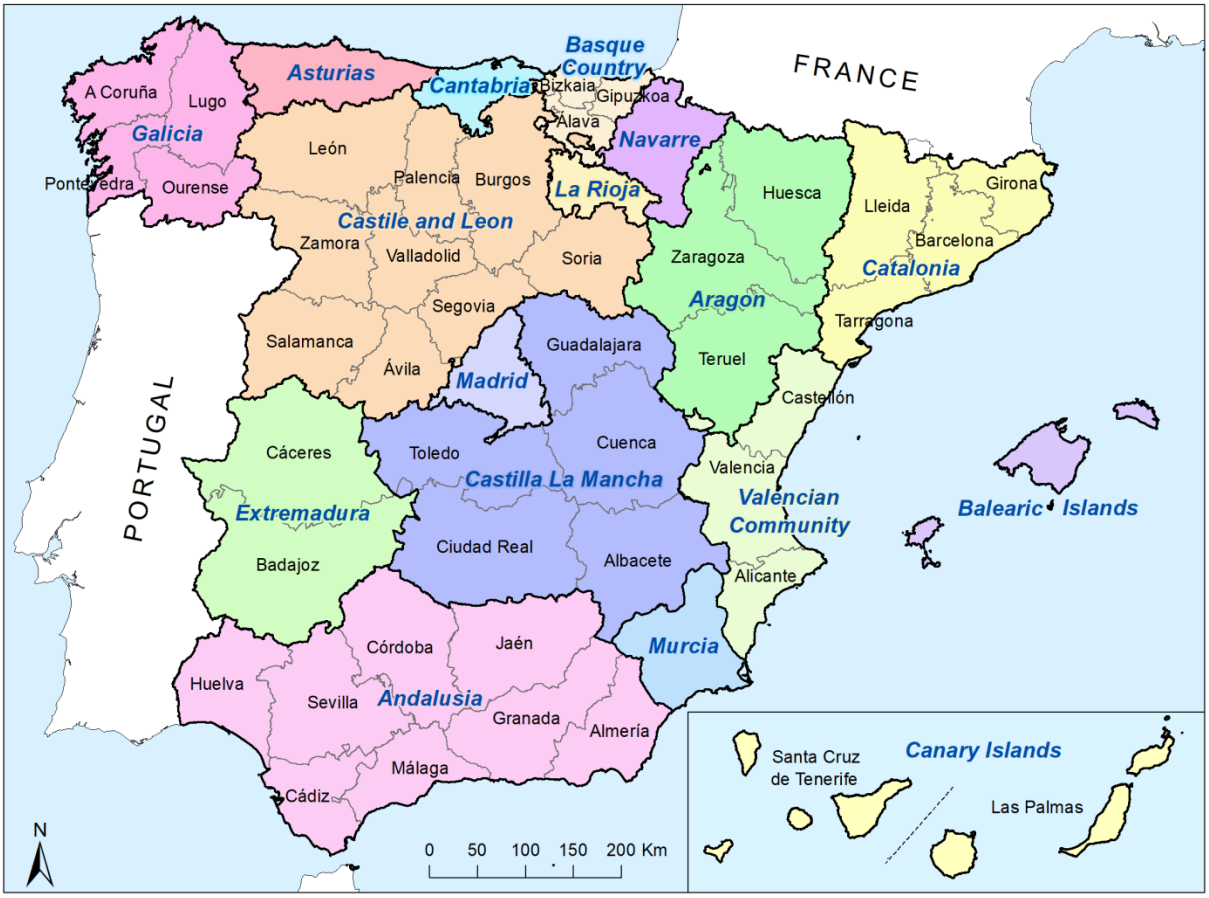

Figure 2. Priority values for on-farm conservation of tomato landraces in Spain according to 16W strategy.

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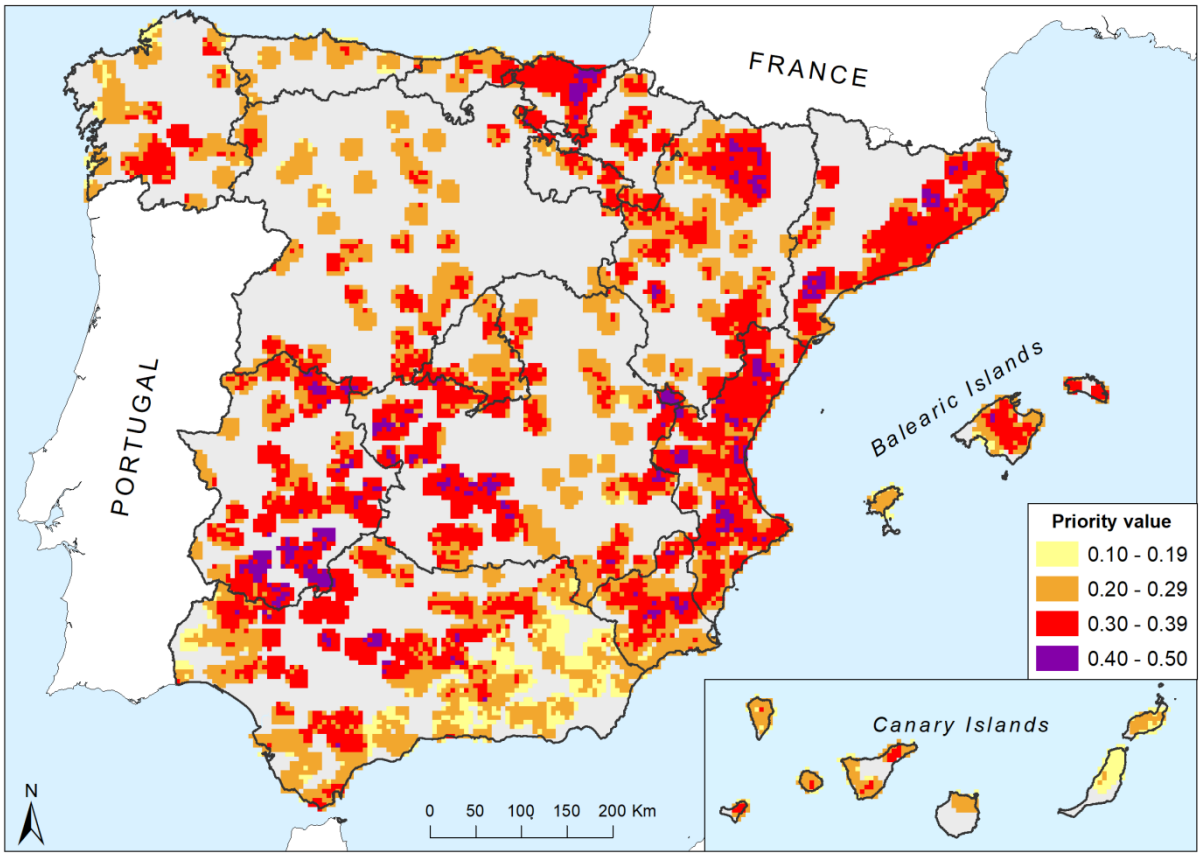

Figure 3. Most appropriate counties for on-farm conservation of tomato landraces in Spain, according to 16W strategy. Color indicates the average priority value of each county, considering the priority values of the area of study within it.

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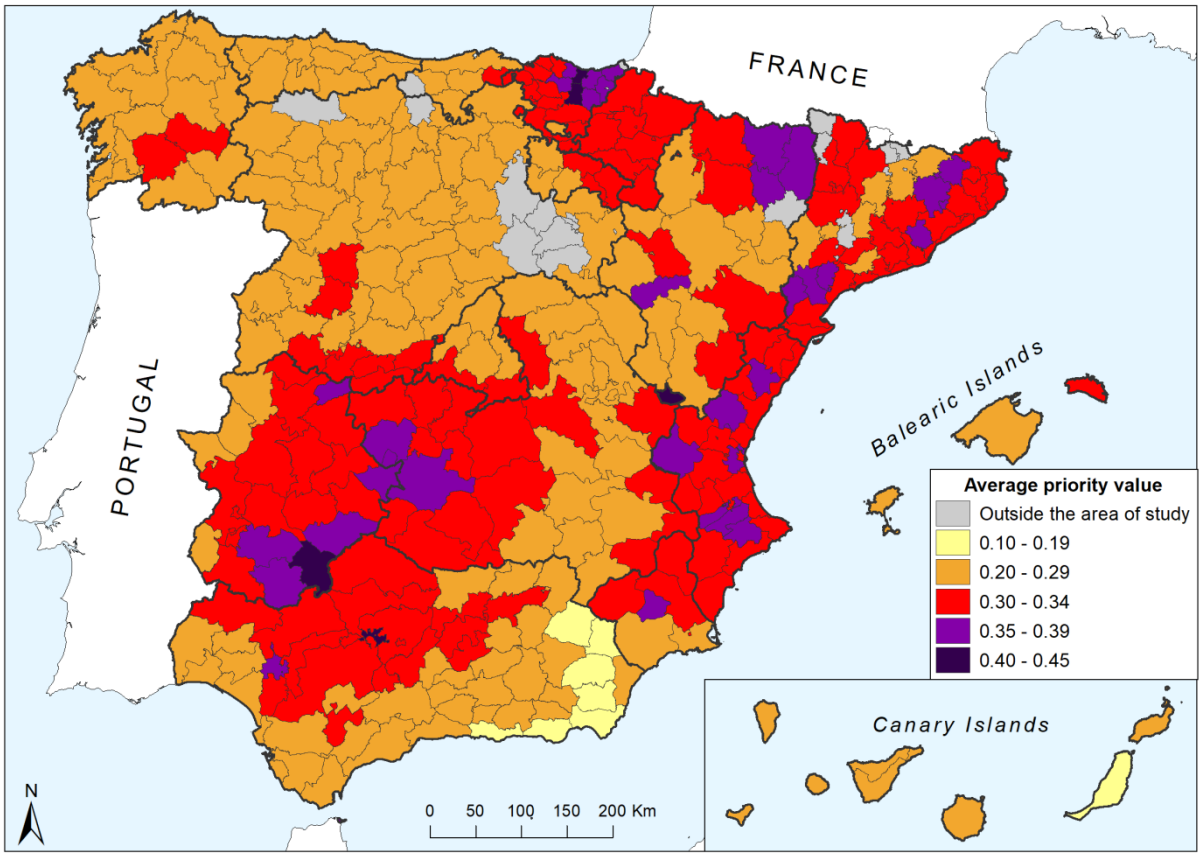

TABLES

Table 1. Criteria subjected to expert evaluation for the identification of the MAPA for on-farm conservation of tomato landraces.

Category	Criterion
Agrobio-diversity	1. Number of tomato landraces grown in the area
	2. Number of species (other than tomato) grown with landraces in the area
	3. Phenotypic diversity of the tomato landraces grown in the area
	4. Number of non-governmental organizations involved in promotion and conservation of agrobiodiversity that work in the area
	5. Number of landraces cultivated in the area included in the Project "*Ark of Taste*" of "*Slow Food*"
	6. Percentage of the area within the action range (100 km radius) of a "*Km0 Restaurant*" of "*Slow Food*"
Agronomy	7. Average area of the horticultural properties
	8. Percentage of the area dedicated to tomato cultivation
	9. Percentage of the area dedicated to irrigated tomato cultivation
	10. Percentage of abandoned crops in the area
Protected agriculture	11. Permission of tomato production with protected geographical indication
Ecological interest and protected areas	12. Area declared Sites of Community Importance, Special Protection Areas for Birds or other protected natural areas
	13. Area declared Biosphere Reserve

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	14. Area declared of Special Importance for the Mediterranean
	15. Area of ecological interest (fallow land, alfalfa, reforested areas, N fixing crops and agroforestry areas)
Ecogeography	16. Vulnerability to climate change
	17. Environmental conditions that favour the cultivation of tomato landraces with potential interesting abiotic adaptations (to high temperature and saline soils)
	18. Ecogeographical diversity
Demography	19. Population density
	20. Population variation (population difference between 1991 and 2016)
	21. Average population age
	22. Percentage of population > 65 years old
	23. Ratio women/men
Socioeconomy	24. Average per capita income
	25. Percentage of active population that is unemployed
	26. Percentage of disadvantaged area
Accesibility	27. Accessibility index by road
	28. Distance to an urban center of more than 10 000 inhabitants

Table 2. Criteria selected for the identification of the MAPA for on-farm conservation of tomato landraces: importance value and favorable status.

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Position in the ranking	Criterion	Weighted average importance value	Favourable status
1	3. Phenotypic diversity of the tomato landraces grown in the area	0.0924	High number
2	1. Number of tomato landraces grown in the area	0.0915	High number
3	2. Number of species (other than tomato) grown with landraces in the area	0.0805	High diversity
4	17. Environmental conditions that favour the cultivation of tomato landraces with potential interesting abiotic adaptations (to high temperature and salinity)	0.0741	High number
5	16. Vulnerability to climate change	0.0631	High number
6	18. Ecogeographical diversity	0.0622	High percentage
7	4. Number of non-governmental organizations involved in promotion and conservation of agrobiodiversity that work in the area	0.0595	Low area
8	22. Percentage of population > 65 years old	0.0567	High percentage
9	6. Percentage of the area within the action range (100 km radius) of a "*Km0 Restaurant*" of "*Slow Food*"	0.0558	Permission
10	8. Percentage of the area dedicated to tomato cultivation	0.0531	High vulnerability
10	11. Permission of tomato production with protected geographical indication	0.0531	With those environmental conditions
11	26. Percentage of disadvantaged area	0.0522	High diversity
11	28. Distance to an urban center of more than 10 000 inhabitants	0.0522	Low average
12	21. Average population age	0.0512	Low percentage
12	5. Number of landraces cultivated in the area included in the Project "*Ark of Taste*" of "*Slow Food*"	0.0512	High percentage
12	7. Average area of the horticultural properties	0.0512	High distance

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